

Fatemeh Bagheri

Space and Planetary Scientist
Astrophysicist

CONTACT

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📍 US citizen

? INTERESTS

- Planetary magnetosphere
- Exoplanets
- Astrobiology and habitability
- Stellar wind-magnetosphere coupling

🎓 EDUCATION

University of Texas at Arlington (UTA)

2021 **Ph.D.**
Planetary and Space Sciences
Dissertation: Energy Dissipation during Geomagnetic Storms

Sharif University of Technology (SUT)

admission rate: Top 1% national
2021 **Ph.D.**
Astrophysics
Dissertation: Detection of Exoplanets using Space-based telescope via microlensing binary-source events

2013 M.Sc. Gravity and Cosmology
2011 B.Sc. Physics

SKILLS

Programming languages

Advanced: Python, Fortran

Intermediate: MATLAB, C++

Other

- Git, genAI, deep learning, NN, Cloud computing, high performance computing
- LaTeX, MS Office
- Linux, Windows, and Mac OS

EXPERIENCE

📅 2024-pres. 📍 NASA Goddard Space Flight Center

NASA Postdoctoral Program Fellow

- Conducted advanced numerical simulations to study space weather phenomena affecting Earth and other planetary environments, contributing to improved understanding of heliophysics and planetary system dynamics.

📅 2023 📍 UTA

Assistant Professor of Instruction

- Taught an undergraduate course on Python programming and data science fundamentals.

📅 2022-2023 📍 UTA

NSF MPS/Ascend Postdoctoral Fellow

- Developed innovative modeling techniques to characterize planetary magnetic environments and their radio-frequency emissions.
- Performed high-fidelity MHD simulations using the GAMERA code to analyze magnetospheric interactions with stellar winds, supporting mission planning, trajectory safety, and environmental risk assessment for space vehicles operating in magnetized planetary systems.
- Utilized the GAMERA MHD code to simulate interactions between exoplanetary magnetic fields and various stellar winds.

📅 2022 📍 UTA

American Physical Society (APS) DSECOP Fellow

- Designed and delivered two educational modules on applied AI techniques—including deep learning and random forest—for undergraduate physics students.

📅 2019-2021 📍 UTA

Graduate research assistant **Advisor: Ramon E. Lopez**

- Performed a large-scale analysis of geomagnetic storm events to evaluate energy transfer in near-Earth space, uncovering key dependencies between solar wind parameters and ionospheric energy dissipation—informing improved space weather impact assessments for spacecraft operations.
- Assessed Joule heating during geomagnetic disturbances by integrating observational data, empirical models, and high-resolution SWMF simulations, identifying critical discrepancies that can influence thermal loading and system performance in LEO environments.

📅 2017-2018 📍 UT Austin

Visiting scholar **Advisor: John Wheeler**

- Investigated the conditions required for primordial RNAs to evolve into the first DNA molecules in oceanic environments.

📅 2011-2017 📍 SUT

Graduate research assistant **Advisor: Sohrab Rahvar**

- Investigated the potential for detecting exoplanets orbiting source stars during microlensing events using Roman and Spitzer telescopes by conducting Monte Carlo simulations of their observations. Analyzed detection probabilities, finding negligible detection chances for rocky planets but a non-zero probability for Jovian planets.
- (Advisor: Reza Mansouri) Discovered an axially symmetric spacetime that leads to a spherically symmetric Einstein tensor, providing a solution of Einstein's equations with a spherically symmetric matter distribution in a non-spherically symmetric spacetime.

AWARDS and FUNDING HISTORY (500K in total)

 2024-2026  NASA Goddard Space Flight Center  200,000
NASA Postdoctoral Program (NPP) Fellowship (PI)

 awarded by: NASA Goddard Space Flight Center

 2023  Argonne National Laboratory
Argonne Training Program on Extreme-Scale Computing (ATPESC)

 awarded by: Argonne National Laboratory

 2023  UT Austin
Rising Stars in Computational & Data Sciences

 awarded by: UT Austin's Oden Institute for Computational Engineering and Sciences,
Sandia National Laboratories, and Lawrence Livermore National Laboratory

 2022-2023  UTA  200,000
NSF- MPS/Ascend Postdoctoral Fellowship (PI)

 awarded by: National Science Foundation

 2022  UTA  5,000
APS- DSECOP Fellowship (PI)

 awarded by: American Physical Society

 2017-2018  UT Austin  20,000
Visiting Scholarship

 awarded by: National Elites Foundation

MENTORSHIP

 2019-2024  UTA
Anshuman Garga - undergraduate - research co-advisor
Co-advising with Ramon Lopez. Radio emission of exoplanets.

Rana Abdulaziz - undergraduate - research co-advisor
Co-advising with Ramon Lopez. Formation of exoplanets.

Rushikesh Patil - undergraduate - research co-advisor
Co-advising with Ramon Lopez. Solar wind-magnetosphere's data analysis.

Chelsi Nelson - undergraduate - research co-advisor
Co-advising with Ramon Lopez. Magnetopause location, analyzing THEMIS data.

 2012 – 2014  Mobina High School, Tehran, Iran
Developed and taught a course with title “Critical Thinking”.
Organized three Scientific Experiments Workshops on Scientific Experiment.

TEACHING ASSISTANT

📅 2017 – 2018	General Physics I Laboratory, Junior Undergraduate, SUT
📅 2015 – 2016	Astronomy/ Analytical Mechanics II, Senior Undergraduate, SUT
📅 2014 – 2015	Analytical Mechanics I II, Senior Undergraduate, SUT
📅 2013 – 2014	Mathematical Physics II, Senior Undergraduate, SUT
📅 2012 – 2013	Electroacoustics, Senior Undergraduate, SUT
📅 2011 – 2012	Mathematical Physics I, Senior Undergraduate, SUT

SERVICE

Panelist/Reviewer, National Science Foundation (NSF), Astronomy Program
Peer Reviewer for scientific journals

MEDIA COVERAGE

Early Career Spotlight at NASA Goddard Space Flight Center – September 2025

Featured in Early Career Spotlight at NASA Goddard Space Flight Center.



Nature's Magnifying Glasses Reveal Distant Planets – APS News, April 2025

Featured in APS News for my presentation and research on detecting exoplanets through gravitational microlensing and magnetic radio emissions.



Physics Doctoral Student Receives NSF Postdoctoral Fellowship – UT Arlington News, December 2021

Coverage of my NSF MPS-Ascend Postdoctoral Fellowship award and research on exoplanet detection using magnetic radio emissions.



ATPESC – Argonne National Laboratory, August 2023

Highlighted among selected participants in the ATPESC summer program, which provides advanced training in high-performance computing for computational scientists.



PUBLICATIONS (13 papers in peer-reviewed journals, 1 preprint on arXiv, 9 first-author papers)

Saturation of Ionospheric Currents Under Extreme Environments

 **F. Bagheri**, A. Glocer

 2025

 Astrophysical Journal (ApJ) (under review)

A New X-line Model: Comparison to MHD Magnetic Separator

 B. de Welle, H. K. Connor, D. G. Sibeck, A. Glocer, S. A. Fuselier, K. Trattner, S. M. Petrinec, A. Brenner, **F. Bagheri**, S. Lee

 2025

 Journal of Geophysical Research-Space Sciences (JGR) (under review)

Impacts of Tidal Locking on Magnetospheric Energy Input to Exoplanet Atmospheres

 **F. Bagheri**, A. Glocer, R E. Lopez

 2025

 Astrophysical Journal (ApJ)



Exploring Radio Emissions from Confirmed Exoplanets Using SKA

 **F. Bagheri**, A. Garga, R E. Lopez

 2024

 Frontiers in Astronomy and Space Science



Infrared-Radio-follow-up Observations for Detection of the Magnetic Radio Emission of Extra Solar Planets: A New Window to Detect Exoplanets

 **F. Bagheri**, A. Shahmoradi, R E. Lopez

 2024

 Frontiers in Astronomy and Space Science



A Fresh Look into the Interaction of Exoplanets Magnetosphere with Stellar Winds using MHD Simulation

 **F. Bagheri**, K. Pham, R E. Lopez

 2024

 Frontiers in Astronomy and Space Science



An alternative statistical interpretation for the first direct evidence of Collapsars

 J A. Osborne, C M. Bryant, **F. Bagheri**, A. Shahmoradi

 2024

 Astronomy & Astrophysics (A&A)



Comparison of Empirical Models of Ionospheric Heating to Global Simulation

 **F. Bagheri**, R E. Lopez

 2023

 Frontiers in Astronomy and Space Science



Solar Wind Magnetosonic Mach Number as a Control Variable for Energy Dissipation during Magnetic Storms

 **F. Bagheri**, R E. Lopez

 2022

 Frontiers in Astronomy and Space Science



Bayesian Methods for Inferring Missing Data in the BATSE Catalog of Short Gamma-Ray

Bursts

 J A. Osborne, **F. Bagheri**, A. Shahmoradi

 2022

 Universe



ParaDRAM: A Cross-Language Toolbox for Parallel High-Performance Delayed-Rejection Adaptive Metropolis Markov Chain Monte Carlo Simulations

 A. Shahmoradi, **F. Bagheri**

 2022

 Computer Methods in Applied Mechanics and Engineering (accepted) 

Are there radio-loud and radio-quiet Gamma-Ray Bursts?

 J A. Osborne, **F. Bagheri**, A. Shahmoradi

 2021

 Monthly Notices of the Royal Astronomical Society (MNRAS)



ParaMonte: A high-performance serial/parallel Monte Carlo simulation library for C, C++, Fortran

 A. Shahmoradi, **F. Bagheri**, J A. Osborne

 2021

 Journal of Open-Source Software (JOSS)



Detection of Exoplanet as a Binary Source of Microlensing Events in WFIRST Survey

 **F. Bagheri**, S. Sadjadian, S. Rahvar

 2019

 Monthly Notices of the Royal Astronomical Society (MNRAS)



The symmetry of the energy momentum tensor does not necessarily reflect the space-time symmetry: a viscous axially symmetric cosmological solution

 **F. Bagheri**, R. Mansouri

 2014

 arXiv



SELECTED TALKS and CONFERENCE PRESENTATIONS



2025



Anaheim, CA

APS Global Physics Summit 2025

Oral Presentation:

Infrared-Radio-follow-up Observations for Detection of the Magnetic Radio Emission of Extra-Solar Planets: A New Window to Detect Exoplanets

Poster Presentation:

Impacts of Tidal Locking on Exoplanets Magnetosphere Energy



2025



National Harbor, MD

245th Meeting of The American Astronomical Society

Oral Presentation:

1. Impacts of Tidal Locking on Exoplanets Magnetosphere Energy

2. Infrared-Radio-follow-up Observations for Detection of the Magnetic Radio Emission of Extra Sola Planets: A New Window to Detect Exoplanets and Exomoons



2024



Washington, D.C.

Fall Meeting of the American Geophysical Union

Poster Presentation: C

Impacts of Tidal Locking on Exoplanets Magnetosphere Energy



2024



Greenbelt, MD

The Roman Galactic Exoplanet Survey Project Infrastructure Team (RGES PIT) Year 2 F2F Meeting

Hosted by NASA Goddard Space Flight Center



2024



Houston, TX

Astrobiology and the Future of Life Meeting

Oral Presentation:

Infrared-Radio-follow-up Observations for Detection of the Magnetic Radio Emission of Extra Sola Planets: A New Window to Detect Exoplanets and Exomoons



2024



Huntsville, AL

4th NASA SMD AI Workshop

Hosted by NASA Headquarter, Astrophysics Division



2024



New Orleans, LA

243rd Meeting of The American Astronomical Society

Oral Presentation:

Infrared-Radio-follow-up Observations for Detection of the Magnetic Radio Emission of Extra Sola Planets: A New Window to Detect Exoplanets and Exomoons

Poster Presentation:

Jovian Exoplanets Magnetosphere Interaction with Stellar winds



2024



Arlington, TX

Introduction to Exoplanet Science II (one-day workshop), UTA

Organizer and Presenter



2023



Pasadena, CA

IPAC, California Institute of Technology

Invited talk:

Magnetic Radio Emission as A New Window to Exoplanets Detection



2023



Austin, TX

Oden Institute for Computational Engineering and Sciences

Invited talk:

Finding Aliens: MA Study of Exoplanets Magnetosphere Interactions with Stellar Winds using MHD Simulations

-  2023  Arlington, TX
Introduction to Exoplanet Science I (one-day workshop), UTA
Organizer and Presenter
-  2023  Arlington, TX
UTA Physics colloquia, UTA
Invited Talk:
New Method to Detect Exoplanets using their Radio Emission
-  2023  Seattle, WA
241st Meeting of The American Astronomical Society
Oral Presentation:
A Study of Exoplanets Magnetosphere Interactions with Stellar Winds using MHD Simulations
-  2022  Pasadena, CA
Sagan Summer Workshop, Caltech
Hosted by the Nasa Exoplanet Science Institute
-  2022  Honolulu, HI
Geospace Environment Modeling (GEM) Summer Workshop
Poster Presentation:
Comparison of Empirical Models of Ionospheric Joule Heating to Global Simulations
-  2022  Los Vegas, NV
Exoplanet IV
Poster Presentation:
Direct Detection of Exoplanets in Microlensing Events
-  2021  New Orleans, LA
Fall Meeting of the American Geophysical Union
Poster Presentation: C
Comparison of Empirical Models of Ionospheric Joule Heating to Global Simulations
-  2021  virtual
GEM Summer Workshop
Poster Presentation:
An overview of ionospheric currents on October 8-9, 2012: A CUSIA Event Study
-  2021  virtual
Fall Meeting of the American Geophysical Union
Poster Presentation:
Correlation between Solar Wind Density and Ionospheric Electrojets during the Main Phase of Magnetic Storms
-  2020  virtual
Fall Meeting of the American Physical Society Texas Section
Oral Presentation:
Energy Dissipation in Storms with Low Solar Wind Magnetosonic Mach Number
-  2020  virtual
GEM Summer Workshop
Poster Presentation:
Correlation between solar wind density and ionospheric electrojets during the main phase of magnetic storms
-  2020  virtual
March Meeting of the American Physical Society
-  2019  San Francisco, CA

Fall Meeting of the American Geophysical Union

Poster Presentation:

Multipoint Observations of Solar Wind Conditions and Magnetopause Motion

📅 2019

📍 Lubbock, TX

Fall Meeting of the American Physical Society Texas Section

Oral Presentation:

Identifying Magnetospheric Crossings between Northward and Southward IMF

📅 2019

📍 Boulder, CO

Boulder Space Weather Summer School (SWSS)

Hosted by National Center for Atmospheric Research

📅 2019

📍 Denver, CO

April Meeting of the American Physical Society

Oral Presentation:

The symmetry of the energy momentum tensor does not necessarily reflect the space-time symmetry:
a viscous axially symmetric cosmological solution

Poster Presentation:

Exoplanet Detection by Spitzer via Microlensing Events

📅 2019

📍 Boston, MA

March Meeting of the American Physical Society

📅 2018

📍 Austin, TX

Understanding of Planetary Habitability-Pop-up institute

Hosted by UT Austin

📅 2018

📍 Los Angeles, CA

March Meeting of the American Physical Society

📅 2016

📍 Trieste, Italy

Cosmology with Next Generation Radio Surveys

Hosted by ICTP

📅 2016

📍 Trieste, Italy

Summer School in Cosmology

Hosted by ICTP

REFERENCES

Ramon E. Lopez, PhD Relationship: PhD/Postdoc Adviser, (2020 – 2023)
Professor of Physics,
University of Texas at Arlington
Member, U.T. System & U.T. Arlington Academy of Distinguished Teachers

Contact Information:

Department of Physics E-mail: relopez@uta.edu
University of Texas at Arlington Phone: 817-272-0386
502 Yates St., Science Hall, Room 008,
Arlington, TX 76019, USA

Alex Glocer, PhD Relationship: Postdoc Adviser, (2024 – present)
Research Astrophysicist,
NASA Goddard Space Flight Center

Contact Information:

Building 021 | Room 220 E-mail: alex.glocer-1@nasa.gov
NASA Goddard Space Flight Center, Phone: 301.286.9475
Greenbelt, MD 20771, USA

David P. Bennett, PhD Relationship: collaborator, (2024 – present)
Sr. Research Scientist,
NASA Goddard Space Flight Center/University of Maryland at College Park

Contact Information:

Building 034 | Room E303 E-mail: bennettd@umd.edu
NASA Goddard Space Flight Center Phone: 301.286.5473
Greenbelt, MD 20771, USA

Yue Deng, PhD Relationship: PhD Professor, (2020 – 2021)
Professor of Physics,
University of Texas at Arlington

Contact Information:

Department of Physics E-mail: yuedeng@uta.edu
University of Texas at Arlington Phone: 817-272-2469
502 Yates St., CPB, Room 334,
Arlington, TX 76019, USA

Zdzislaw E. Musielak, PhD Relationship: PhD Professor, (2020 – 2021)
Professor of Physics,
University of Texas at Arlington

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